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A TECHNICAL SEMINAR REPORT ON

CLOUD COMPUTING:

AMAZON WEB SERVICES, ELASTIC

COMPUTE CLOUD,

IDENTITY AND ACCESS MANAGEMENT,

AND ELASTIC LOAD BALANCING

BACHELOR OF ENGINEERING IN

INFORMATION SCIENCE & ENGINEERING

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ABSTRACT

Cloud computing has revolutionized how organizations architect, deploy, and maintain IT infrastructure. By providing on-demand access to computing resources via the Internet, cloud platforms eliminate the necessity for expensive on-premises hardware while delivering virtually limitless scalability. This report presents a comprehensive study of cloud computing principles and their practical implementation on Amazon Web Services (AWS) — the world's foremost cloud platform. Topics explored include the fundamental characteristics of cloud computing, service models, an overview of AWS infrastructure, launching Amazon Elastic Compute Cloud instances, configuring web servers, managing access through Identity and Access Management, and implementing Elastic Load Balancers for high availability. This report serves as a practical guide for students and professionals seeking to master modern cloud infrastructure.

Keywords: Cloud Computing, Amazon Web Services, Elastic Compute Cloud, Identity and Access Management, Elastic Load Balancing, Scalability.

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1. Introduction

The exponential growth of internet connectivity and increasing demand for flexible, scalable computing resources have established cloud computing as the dominant paradigm in modern IT. Rather than investing capital in physical hardware, organizations can now lease computing resources on a pay-as-you-go basis from cloud providers. This shift from Capital Expenditure (CapEx) to Operational Expenditure (OpEx) has democratized access to enterprise-grade infrastructure, enabling both startups and large corporations to innovate faster.

Cloud computing delivers three primary service categories: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). These services are accessible through public, private, or hybrid deployment models. Amazon Web Services (AWS), launched in 2006, pioneered the cloud industry and remains the market leader, offering over 200 comprehensive services spanning compute, storage, databases, networking, machine learning, and more.

This report examines the following key aspects of AWS:

- Understanding Cloud Computing and its core characteristics
- Cloud Service and Deployment Models
- Overview of Amazon Web Services
- Launching Amazon Elastic Compute Cloud (EC2) Instances
- Deploying Web Servers on EC2
- Identity and Access Management (IAM)
- Elastic Load Balancing (ELB)

2. Understanding Cloud Computing

Cloud computing is defined by the National Institute of Standards and Technology (NIST) as *“a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.”*

2.1 Essential Characteristics

- **On-Demand Self-Service:** Users can independently provision resources automatically without requiring human interaction with the service provider.
- **Broad Network Access:** Resources are accessible over the network through standard mechanisms across heterogeneous platforms (laptops, tablets, smartphones).
- **Resource Pooling:** Provider resources are pooled to serve multiple consumers using a multi-tenant model with dynamic resource assignment.
- **Rapid Elasticity:** Capabilities can be elastically provisioned and released to scale rapidly with demand, often appearing unlimited to consumers.
- **Measured Service:** Resource usage is automatically monitored, controlled, and reported, providing transparency for both provider and consumer.

2.2 Cloud Service Models

- **Infrastructure as a Service (IaaS):** Provides fundamental computing resources including virtual machines, storage, and networking. Users manage the OS and applications while the provider handles physical infrastructure. *Example: Amazon EC2.*
- **Platform as a Service (PaaS):** Offers a managed development environment. Users focus on application code while the provider manages infrastructure and runtime. *Example: Google App Engine, AWS Elastic Beanstalk.*
- **Software as a Service (SaaS):** Delivers fully functional applications over the internet without installation or maintenance requirements. *Example: Gmail, Microsoft 365, Salesforce.*



Figure 1: Cloud Computing Service Models

3. Amazon Web Services (AWS)

Amazon Web Services (AWS) is the world's most comprehensive and widely adopted cloud platform offered by Amazon.com. Launched commercially in 2006, AWS provides on-demand cloud computing services to individuals, enterprises, and governments on a pay-as-you-go pricing model. With data centers across 33 geographic regions and over 105 Availability Zones worldwide, AWS delivers low-latency, highly available services on a global scale.

3.1 AWS Global Infrastructure

AWS infrastructure is organized around the following key concepts:

- **Regions:** Physical geographical locations where AWS clusters data centers. Each Region is completely independent (e.g., `us-east-1`, `ap-south-1`).
- **Availability Zones (AZs):** Isolated locations within a Region, each consisting of one or more data centers with redundant power, networking, and connectivity. Applications deployed across multiple AZs achieve high availability.
- **Edge Locations:** Endpoints used by Amazon CloudFront (CDN) to cache content closer to end users for low-latency delivery.

3.2 Core AWS Services

Service Category	Key Services
Compute	EC2, Lambda, Elastic Beanstalk, ECS
Storage	S3, EBS, Glacier, EFS
Database	RDS, DynamoDB, Aurora, Redshift
Networking	VPC, Route 53, CloudFront, ELB
Security	IAM, KMS, Shield, WAF
Monitoring	CloudWatch, CloudTrail, Config



Figure 2: AWS Global Infrastructure

4. Amazon Elastic Compute Cloud (EC2)

Amazon Elastic Compute Cloud (EC2) is AWS's core compute service providing resizable virtual servers (called *instances*) in the cloud. EC2 enables users to launch and manage virtual machines without investing in physical hardware, allowing organizations to scale capacity within minutes based on demand.

4.1 Key EC2 Concepts

- **Amazon Machine Image (AMI):** A pre-configured template containing the operating system and software required to launch an instance (e.g., Amazon Linux 2023, Ubuntu 22.04).
- **Instance Types:** Define the hardware specifications including CPU, memory, storage,

and network capacity. Examples: `t2.micro` (free tier), `m5.large`, `c5.xlarge`.

- **Key Pairs:** Used for secure SSH access to Linux instances or RDP access to Windows. Consists of a public key (stored by AWS) and private key file (stored by user).
- **Security Groups:** Virtual firewalls controlling inbound and outbound traffic to instances.
- **Elastic IP:** A static public IPv4 address that can be associated with instances.

4.2 Steps to Launch an EC2 Instance

1. Log in to the AWS Management Console and navigate to **EC2**.
2. Click **Launch Instance** and provide a descriptive name.
3. Select an **AMI** (e.g., *Amazon Linux 2023*).
4. Choose an **Instance Type** (e.g., `t2.micro` for free tier).
5. Create or select an existing **Key Pair** for SSH access.
6. Configure **Network** settings and **Security Group** (allow port 22 for SSH, port 80 for HTTP).
7. Configure **Storage** (default 8 GiB EBS volume is sufficient for basic use).
8. Review the configuration and click **Launch Instance**.
9. After launch, copy the **Public IP** and connect via SSH:

```
ssh -i "your-key.pem" ec2-user@<Public-IP>
```

5. Deploying a Web Server on EC2

Once an EC2 instance is operational, it can be configured as a fully functional web server using the Apache HTTP Server (`httpd`). This demonstrates how cloud virtual machines can host websites and applications accessible over the public internet.

5.1 Installing Apache on Amazon Linux

After connecting to the EC2 instance via SSH, execute the following commands:


```
# Update all packages
sudo yum update -y

# Install Apache web server
sudo yum install httpd -y

# Start and enable Apache
sudo systemctl start httpd
sudo systemctl enable httpd

# Create a simple HTML page
echo "<h1>Welcome to AWS EC2 Web Server!</h1>" | sudo tee /var/www/html
```

5.2 Verifying the Web Server

1. Ensure the Security Group attached to the instance allows **inbound HTTP traffic on port 80**.
2. Open a browser and navigate to `http://<Public-IP>`.
3. The page should display “*Welcome to AWS EC2 Web Server!*” confirming successful deployment.

6. Identity and Access Management (IAM)

AWS Identity and Access Management (IAM) is a web service that enables organizations to securely control access to AWS services and resources. IAM allows administrators to manage *who* (authentication) can perform *what actions* (authorization) on *which resources* in the AWS environment.

6.1 Core IAM Components

- **Users:** Individual identities with unique security credentials including username, password, and optionally access keys for programmatic access.
- **Groups:** Collections of users that share common permissions. Permissions assigned to groups are inherited by all member users.

- **Roles:** Identities with specific permissions that can be assumed by AWS services, applications, or federated users. Roles provide temporary credentials rather than permanent ones.
- **Policies:** JSON documents that define permissions, specifying which actions are allowed or denied on which resources.

6.2 Creating an IAM Administrator User

The AWS root account should never be used for routine operations. Instead, create an IAM user with administrative privileges:

1. Navigate to **IAM** in the AWS Console.
2. Click **Users** → **Create User**.
3. Enter a username (e.g., `admin-user`) and enable **Console Access**.
4. Attach the **AdministratorAccess** managed policy directly or via a group.
5. Complete the wizard and download the **credentials CSV** for safekeeping.
6. Enable **Multi-Factor Authentication (MFA)** for enhanced security.

6.3 IAM Best Practices

- Follow the **principle of least privilege** — grant only necessary permissions.
- Never share or embed root account credentials in code or configuration files.
- Use **IAM Roles** for EC2 instances instead of embedding access keys.
- Regularly **rotate credentials** and review policies using Access Advisor.
- Enable **CloudTrail** to audit all API calls across the account.
- Implement strong password policies and require MFA for privileged users.

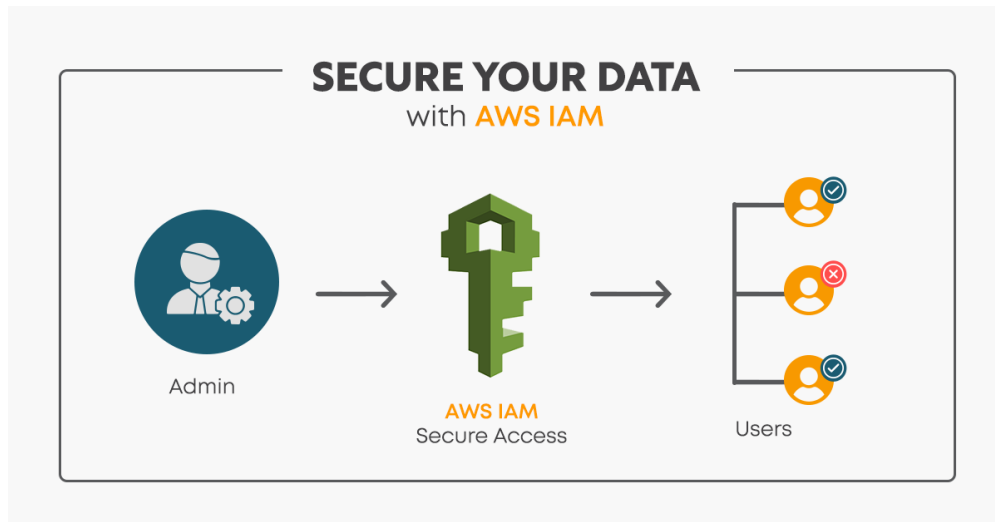


Figure 3: IAM Administrator User Configuration

7. Elastic Load Balancing (ELB)

Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple targets such as EC2 instances, containers, IP addresses, or Lambda functions. Load balancers enhance application availability and fault tolerance by ensuring no single resource becomes overwhelmed with requests.

7.1 Types of AWS Load Balancers

- **Application Load Balancer (ALB):** Operates at Layer 7 (HTTP/HTTPS). Supports path-based and host-based routing, ideal for web applications and microservices.
- **Network Load Balancer (NLB):** Operates at Layer 4 (TCP/UDP). Handles millions of requests per second with ultra-low latency.
- **Gateway Load Balancer (GWLb):** Used for deploying and scaling virtual network appliances such as firewalls and intrusion detection systems.

7.2 Setting Up an Application Load Balancer

1. Navigate to **EC2** → **Load Balancers** → **Create Load Balancer**.
2. Select **Application Load Balancer**.
3. Provide a name and select **Internet-facing** scheme.
4. Choose at least **two Availability Zones** and their corresponding subnets.

5. Assign a **Security Group** that allows inbound HTTP (port 80) traffic.
6. Create a **Target Group**: select *Instances* as target type and HTTP as protocol.
7. Register the running **EC2 instances** to the target group.
8. Complete the wizard and click **Create Load Balancer**.
9. Once active, copy the **DNS name** of the ALB and test it in a browser — traffic will be distributed across registered instances.

7.3 Health Checks

The ALB periodically sends health check requests to registered targets (e.g., GET / on port 80). Instances that fail health checks are automatically removed from rotation until they recover, ensuring continuous availability.



Figure 4: Application Load Balancer Architecture